Monte Carlo Methods:
Concepts and Computational Foundations

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guide

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This is a collection of notes based on the Monte Carlo topics from my Mathematical Methods for Physicists II (MMP2) coursework. It provides undergraduate foundations for notebook 3A and the Monte Carlo sampling used in the later guided Ising project, 3B. Earlier probability and random-walk material is referenced where useful. These notes explain the logic of the calculations and the limits of their statistical interpretation; they are a learning companion rather than a substitute for a textbook.

1 What Monte Carlo is doing

Monte Carlo methods estimate quantities using random samples. The randomness belongs to the *calculation*; the quantity being estimated can be completely deterministic, such as π or a definite integral. In statistical physics, random sampling can also represent the probabilities of microscopic configurations at thermal equilibrium.

In 3A, independent points estimate area fractions and averages of functions. In 3B, a Markov chain samples interacting-spin configurations and estimates their observables. The common idea is to replace an expectation by a sample average. The important difference is whether successive samples are independent and how the target distribution is generated.

The earlier 1A/1B notes introduce probability, means, variance, random walks, and diffusion. We recall only the ingredients needed here and develop the new Monte Carlo logic. The separate 3A guide follows the exercises; the 3B guide develops Ising physics and then applies this computational foundation.

2 Random variables, expectations, and sample averages

A random variable is a numerical quantity determined by a random outcome. For discrete values x_j with probabilities p_j,

$$\mathbb{E}[X] = \sum_j x_j p_j.$$

For a continuous variable with probability density $p(x)$,

$$\mathbb{E}[f(X)] = \int f(x)p(x)\,dx, \qquad \int p(x)\,dx = 1.$$

The expectation is a property of a distribution. An observed sample mean is a random estimate of it:

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If the X_i have the same mean μ, then $\mathbb{E}[\overline{X}_n] = \mu$. If they are also independent with variance σ^2,

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}.$$

The averaging reduces variance because independent fluctuations do not all reinforce one another. Correlation changes this formula; it cannot be ignored for 3B.

3 Hit-or-miss sampling: turn geometry into probability

Let D be a region inside a box of known area A_B. Choose a point uniformly in the box and define its indicator

$$H = \begin{cases} 1, & \text{inside } D, \\ 0, & \text{outside } D. \end{cases}$$

Uniform sampling means that equal-area parts of the box have equal probability. Thus $p = \mathbb{P}(H = 1) = A_D/A_B$. For n independent points, the hit fraction $\widehat{p} = n_{\mathrm{hit}}/n$ estimates p, and

$$\widehat{A}_D = A_B\widehat{p}.$$

The unit disk in a square of area four gives $\widehat{\pi} = 4\widehat{p}$. An ellipse with semiaxes two and one inside a rectangle of area eight gives $\widehat{A} = 8\widehat{p}$.

The number of hits has a binomial distribution. Since $H^2 = H$,

$$\mathbb{E}[H] = p, \qquad \mathrm{Var}(H) = p - p^2 = p(1-p).$$

Consequently

$$\mathbb{E}[\widehat{A}_D] = A_D, \quad \mathrm{Var}(\widehat{A}_D) = \frac{A_B^2 p(1-p)}{n}.$$

This assumes continuous uniform sampling and independent draws. A nonuniform point cloud generally cannot use an unweighted hit fraction as an area ratio.

4 Uncertainty: variance, spread, and standard error

4.1 The three quantities

Variance measures mean squared deviation and has squared units. Standard deviation is its square root and describes the spread of a distribution. Standard error describes the spread of a particular *estimator*, usually a mean.

If R independent runs return estimates $a_1, \ldots, a_R$, calculate

$$\bar{a} = \frac{1}{R}\sum_r a_r, \qquad s_a^2 = \frac{1}{R-1}\sum_r (a_r - \bar{a})^2, \qquad \widehat{\mathrm{SE}}(\bar{a}) = s_a/\sqrt{R}.$$

Here s_a describes a single run's spread, while $s_a/\sqrt{R}$ describes uncertainty in their average. They answer different questions. The $R-1$ denominator compensates for estimating the mean from the same observations; in NumPy it corresponds to `ddof=1`.

4.2 Two levels of averaging

For 50 independent estimates of π, each using 10,000 points, the one-run uncertainty concerns 10,000 points. The average uses a total of 500,000 independent points and has a smaller uncertainty. Equal-sized batches make the average of the estimates equal to the estimator formed from all the hits pooled together.

Do not use the variance itself as an error bar in the same units as area or an integral. Also do not confuse the width of a histogram of run estimates with the much smaller error bar on its mean.

5 Why larger samples usually help

The law of large numbers explains convergence of independent sample averages under appropriate integrability conditions. The central limit theorem says that for finite-variance independent identically distributed samples, the normalized fluctuation of their mean approaches a normal distribution as the sample grows.

A typical standard error scales as $n^{-1/2}$. A tenfold error reduction therefore needs about 100 times as many samples. This is a statistical scaling law, not a promise that every individual larger sample is closer to the answer. Fresh runs at different n can fluctuate in either direction.

With a sufficiently normal sampling distribution and a reliable standard-error estimate, an approximate 95% interval is $\widehat{A} \pm 1.96\,\widehat{\mathrm{SE}}$. It is not a deterministic bound and does not include systematic bias. Small samples, extreme hit probabilities, or strong correlations need more care.

For π, the continuous-square benchmark is

$$\mathrm{SD}(\widehat{\pi}) = 4\sqrt{\frac{(\pi/4)(1-\pi/4)}{n}}.$$

Use an estimated hit probability if the exact probability is not known. In this learning problem the known π supplies a reference, not information needed to define the estimator.

6 Bias, RMS error, and the notebook's fine grid

An estimator's bias is $b = \mathbb{E}[\widehat{A}] - A$. Its mean squared error separates into

$$\mathbb{E}[(\widehat{A} - A)^2] = \mathrm{Var}(\widehat{A}) + b^2.$$

The root of this quantity is RMS error. It measures both systematic offset and sampling spread.

In 3A the area exercises draw integers and divide by 1000. They sample a discrete grid, not all real coordinates in the box. The expected hit fraction is the fraction of grid sites inside the shape. This approximates the continuous geometric fraction, introducing a small discretization bias.

Increasing the number of draws at fixed grid spacing reduces sampling variance but not this bias. The RMS error can eventually approach a bias floor instead of continuing along an $n^{-1/2}$ curve. The circle grid used here has expected $\hat{\pi} = 3.141547$, an offset of about -4.57×10^{-5} from π. This is much smaller than the typical one-run spread at the displayed sample sizes, but matters to the interpretation of asymptotic convergence.

Continuous uniform coordinates would remove this particular grid approximation. That would be a sampling-model change, not simply another random seed. The reviewed coursework notebook retains its original fine-grid sampling with this distinction explained.

7 Monte Carlo integration: an integral is an expectation

For X uniform on $[a, b]$, its density is $1/(b-a)$, so

$$\mathbb{E}[f(X)] = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

Therefore an integral can be estimated without an evenly spaced quadrature grid:

$$I = \int_a^b f(x)\, dx, \qquad \widehat{I} = (b-a)\frac{1}{n}\sum_{i=1}^{n} f(X_i).$$

For independent draws with finite variance,

$$\operatorname{Var}(\widehat{I}) = \frac{(b-a)^2 \operatorname{Var}(f(X))}{n}, \quad \widehat{\operatorname{SE}}(\widehat{I}) = \frac{(b-a)s_f}{\sqrt{n}}.$$

This requires $b > a$ for the stated positive-width form. Units also follow correctly: function units multiplied by coordinate units give integral units.

For $f(x) = \sin(100x)$ on $[0, 2\pi]$, the exact signed integral is zero. Positive and negative contributions cancel in the expectation, but a finite random sample generally does not cancel perfectly. The uniform-sample variance is $1/2$, so theoretical standard error is $2\pi/\sqrt{2n}$. This is about 0.0140 at $n = 100000$.

Replacing f with $|f|$ computes a different integral. Do not confuse a signed integral with geometric area between a curve and an axis. The second integrand in 3A, $|\sin(100x)|^x$, is also a separate function and does not share the zero benchmark.

8 Random numbers, reproducibility, and experiment design

Pseudorandom generators produce deterministic streams that approximate the required sampling distributions. A fixed seed makes a run reproducible; it does not remove Monte Carlo uncertainty. Seeding once before the experiment is useful. Resetting the same seed for every repeated trial reproduces the same batch and destroys the intended independent-repetition experiment.

In 3A NumPy supplies the random stream. In 3B NumPy initializes spins while Python's `random` selects sites and acceptance draws; both streams are seeded. Changing a seed produces another realization, not a systematic accuracy improvement.

Decide what is being held fixed before comparing runs: samples per estimate, repetitions, physical parameters, and the estimator itself. Repeated n values can legitimately denote fresh trials. A connected line through independent estimates should not be mistaken for the history of one growing sample.

9 From independent sampling to importance sampling

The uniform-sampling estimator averages under a simple distribution. For a more general density $q(x) > 0$ wherever $f(x)$ contributes,

$$\int f(x)\, dx = \mathbb{E}_q\left[\frac{f(X)}{q(X)}\right].$$

Choosing q to sample important contributions more often is importance sampling. Its variance depends on the choice and on the existence of the required moments. This weighted-integral formula is conceptual background, not implemented in 3A.

In a statistical-physics configuration space, the desired expectation is already defined under nonuniform thermal probabilities. Uniformly drawing all configurations can be wasteful because many have negligible thermal weight. A Markov chain can instead visit configurations according to the target weights and average the observable directly. If the chain truly samples the desired equilibrium distribution, no extra Boltzmann factor should be multiplied into each measured observable.

10 Markov chains and the target distribution

A Markov chain is a sequence of states whose next-state distribution depends on the current state. It can revisit the same state, including when a proposed update is rejected. A transition probability $P(c \to c')$ defines how configurations change.

A target distribution $\pi(c)$ is stationary if applying the transition leaves it unchanged. Detailed balance,

$$\pi(c)P(c \to c') = \pi(c')P(c' \to c),$$

is a sufficient way to enforce stationarity. It balances probability flow between each pair of states. It does not by itself guarantee that a finite simulation has reached equilibrium or explored every relevant region.

For equilibrium statistical mechanics, the canonical weights are

$$\pi(c) = \frac{e^{-\beta E(c)}}{Z}, \qquad \beta = (k_B T)^{-1}, \quad Z = \sum_c e^{-\beta E(c)}.$$

The Ising guide explains this physical distribution. Computationally the useful fact is that ratios of weights do not require Z:

$$\frac{\pi(c')}{\pi(c)} = e^{-\beta \Delta E}.$$

11 Metropolis sampling and rejected proposals

If the proposal probabilities for a forward and reverse move are symmetric, choose acceptance

$$A(c \to c') = \min(1, e^{-\beta \Delta E}).$$

Energy-lowering and neutral moves are accepted. For an energy increase, compare a uniform random draw $u \in [0, 1)$ with $e^{-\beta \Delta E}$. This allows occasional unfavorable moves; forbidding them would minimize energy rather than sample finite-temperature equilibrium.

With asymmetric proposals the corresponding proposal-probability ratio must also enter the acceptance rule; the simple formula is not universal. Single-spin flips at uniformly chosen sites in 3B are symmetric proposals.

A rejected proposal is still one step of the chain. Record observables at scheduled times whether or not a move was accepted. Measuring only accepted configurations generally changes the sampled distribution. At positive temperature all finite-energy spin flips have nonzero acceptance probability; nevertheless visits between strongly ordered sectors can be extremely rare.

12 Equilibration, correlation, and effective sample size

12.1 Discarding the initial transient

Starting from a convenient state does not mean starting from equilibrium. Discard a warmup portion, then measure. No universal number of sweeps guarantees equilibration. Compare histories or runs with different starts and extend the warmup when relevant; a transient can persist much longer near a transition or between separated sectors.

12.2 Correlated measurements

Let A_i denote successive equilibrium measurements and let $\rho(k)$ be their normalized lag-k autocorrelation. The variance of an average contains covariance terms. For a long stationary series, when these correlations are summable,

$$\mathrm{Var}(\overline{A}) \approx \frac{\sigma_A^2}{n}\left[1 + 2\sum_{k=1}^{\infty}\rho(k)\right].$$

Define $\tau_{\mathrm{int}} = \frac{1}{2} + \sum_{k\geq 1}\rho(k)$ in units of the recording interval. Then $n_{\mathrm{eff}} \approx n/(2\tau_{\mathrm{int}})$. Positive correlations make the effective information smaller than the raw measurement count. Different observables can have different correlation times.

Recording every tenth sweep may reduce correlations but cannot guarantee independence. In 3B, 600 measurement sweeps with a gap of ten give 60 records, not 600 independent samples. Ordinary $s/\sqrt{60}$ errors can still underestimate uncertainty.

12.3 Practical uncertainty methods

One approach is independent, adequately equilibrated runs. Another is block averaging: divide a long stationary record into contiguous blocks, calculate each block mean, and choose blocks long enough that neighboring means are approximately independent. With B suitable blocks, the standard deviation of their means divided by $\sqrt{B}$ estimates uncertainty in the overall mean. Check how the estimate changes with block length, retaining enough blocks to estimate it reliably.

Neither blocking nor thinning repairs a chain that never reaches or explores the required equilibrium distribution. Correct error bars and adequate equilibrium sampling are separate requirements. The notebooks provide introductory measurements without implementing an autocorrelation or blocking analysis.

13 From a question to a workable algorithm

Question	Sampling and estimator	Main checks
Area in a box	Uniform points; box area times hit fraction	Uniformity, boundary test, repeated spread, grid bias
Definite integral	Independent uniform X; width times mean $f(X)$	Signed integrand, finite variance, reference where known
Thermal observable	Equilibrium Markov configurations; mean observable	Energy updates, acceptance rule, warmup, mixing, correlation

Write the target distribution and observable first. Then decide how to generate samples, what to record, how to estimate uncertainty, and what independent reference or limit can validate the result. A smooth-looking curve does not replace these checks.